Chapter 1. Real Numbers and Limits of Sequences

Field and order axioms, completeness, limits, monotone sequences, limsup and liminf, countability

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Analysis rests on a precise description of the real numbers. We take $\mathbb{R}$ as given by a short list of axioms — the field axioms, the order axioms, and the completeness axiom — and derive everything else from them. (A construction of $\mathbb{R}$ satisfying the axioms is sketched in Appendix A.)

1.1 Field and order axioms

Definition (Ordered field)

A **field** is a set F with operations $+$ and $\cdot$ such that both are associative and commutative, $\cdot$ distributes over $+$, there are identities $0 \neq 1$, every a has an additive inverse $-a$, and every $a \neq 0$ has a multiplicative inverse a^{-1}. A field is **ordered** if there is a subset $P \subset F$ (the positive elements) with

(O1) $a, b \in P \Rightarrow a + b \in P$; (O2) $a, b \in P \Rightarrow ab \in P$;

(O3) for each a, exactly one of $a \in P$, $a = 0$, $-a \in P$ holds (trichotomy).

We write $a < b$ if $b - a \in P$.

Proposition 1.1

In an ordered field: (a) $a < b$ and $b < c$ imply $a < c$; (b) $b < c \iff a + b < a + c$; (c) $a > 0$, $b < c$ imply $ab < ac$, while $a < 0$, $b < c$ imply $ab > ac$; (d) $a^2 > 0$ for $a \neq 0$, in particular $1 > 0$; (e) $0 < a < b \Rightarrow 0 < 1/b < 1/a$; (f) for $a, b > 0$, $a < b \iff a^2 < b^2$.

Proof. (a) $c - a = (c - b) + (b - a) \in P$ by (O1). (b) $(a + c) - (a + b) = c - b$. (c) $ac - ab = a(c - b)$; use (O2), and for $a < 0$ note $a > 0 \iff -a < 0$ (trichotomy). (d) If $a > 0$, $a^2 \in P$ by (O2); if $a < 0$, $a^2 = (-a)^2 \in P$. (e) $1/a > 0$ since otherwise $1 = a \cdot (1/a) \leq 0$; then multiply $a < b$ by $1/(ab) > 0$. (f) $b^2 - a^2 = (b - a)(b + a)$ and $b + a > 0$. ∎

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields; $\mathbb{C}$ is not: if $P \subset \mathbb{C}$ satisfied (O1)–(O3), then $-1 = i^2 \in P$ and $1 = 1^2 \in P$, contradicting trichotomy. No finite field is ordered either, since $1 + 1 + \cdots + 1 > 0$ forces all these sums to be distinct.

The **absolute value** is $|a| = a$ for $a \geq 0$ and $-a$ for $a < 0$.

Proposition 1.2

$|a| \geq 0$ with equality iff $a = 0$; $|ab| = |a||b|$; $|a + b| \leq |a| + |b|$; and $\big||a| - |b|\big| \leq |a - b|$.

Proof. The first two are case checks. Since $-|a| \leq a \leq |a|$ and likewise for b, $-(|a| + |b|) \leq a + b \leq |a| + |b|$, which is the triangle inequality. Applying it to $a = (a - b) + b$ gives $|a| - |b| \leq |a - b|$, and symmetrically. ∎

1.2 The completeness axiom

A set $S \subset \mathbb{R}$ is **bounded above** if some u (an *upper bound*) satisfies $s \leq u$ for all $s \in S$. A **least upper bound** or **supremum** $\sup S$ is an upper bound that is $\leq$ every upper bound; $\inf S$ is defined dually.

Definition (Completeness axiom)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Equivalently, every nonempty set bounded below has an infimum: if S is bounded below, the set L of its lower bounds is nonempty and bounded above (by any element of S), and $\sup L = \inf S$. The ordered field $\mathbb{Q}$ is *not* complete: $\{r \in \mathbb{Q} : r^2 < 2\}$ has no rational supremum (Proposition 1.5 and the irrationality of $\sqrt{2}$).

A useful reformulation: $u = \sup S$ iff u is an upper bound and for every $\epsilon > 0$ there is $s \in S$ with $s > u - \epsilon$.

Theorem 1.3 (Archimedean property)

$\mathbb{N}$ is not bounded above. Consequently, for $a > 0$ and any $b \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $na > b$; in particular $1/n \to 0$ in the sense that for every $\epsilon > 0$ some n has $1/n < \epsilon$.

Proof. If $\mathbb{N}$ were bounded above, let $u = \sup \mathbb{N}$. Then $u - 1$ is not an upper bound, so some $n > u - 1$, and $n + 1 \in \mathbb{N}$ exceeds u: contradiction. For the second claim pick $n > b/a$. ■

Theorem 1.4 (Density of the rationals)

If $a < b$ are real, there is $r \in \mathbb{Q}$ with $a < r < b$. The irrationals are dense as well.

Proof. Choose n with $n(b - a) > 1$ and let m be the least integer with $m > na$ (it exists by the Archimedean property and well-ordering). Then $m - 1 \le na$, so $na < m \le na + 1 < nb$ and $r = m/n$ works. For irrationals apply this to $a/\sqrt{2} < b/\sqrt{2}$ and multiply by $\sqrt{2}$. ■

Proposition 1.5

For each $a > 0$ there is a unique $b > 0$ with $b^2 = a$.

Proof. Uniqueness follows from Proposition 1.1(f). Let $S = \{x > 0 : x^2 < a\}$, nonempty (it contains $\min(1, a)/2$) and bounded above by $\max(1, a)$; put $b = \sup S$. If $b^2 < a$, then for small $0 < h < 1$, $(b + h)^2 \le b^2 + h(2b + 1) < a$, contradicting that b is an upper bound. If $b^2 > a$, then for small $h > 0$, $(b - h)^2 \ge b^2 - 2bh > a$, so $b - h$ would be a smaller upper bound. Hence $b^2 = a$. ■

The completeness axiom pins $\mathbb{R}$ down: any two complete ordered fields F, G are isomorphic, i.e. there is a bijection $\phi : F \to G$ preserving sums, products and order (one maps $\mathbb{N} \subset F$ to $\mathbb{N} \subset G$, extends to $\mathbb{Q}$, and then sends x to $\sup\{\phi(r) : r \in \mathbb{Q}, r < x\}$). We also record the arithmetic of suprema: for nonempty bounded S, T, $\sup(S + T) = \sup S + \sup T$, and $\sup(ST) = \sup S \cdot \sup T$ if $S, T \subset (0, \infty)$; here $S + T = \{s + t\}$ and $ST = \{st\}$.

1.3 Limits of sequences

Definition (Convergence)

A sequence (x_n) **converges** to x, written $x_n \to x$ or $\lim x_n = x$, if for every $\epsilon > 0$ there is N such that $n \ge N \Rightarrow |x_n - x| < \epsilon$.

Proposition 1.6

(a) A sequence has at most one limit. (b) A convergent sequence is bounded.

Proof. (a) If $x_n \to x$ and $x_n \to y$ with $x \ne y$, take $\epsilon = |x - y|/2$; for large n, $|x - y| \le |x - x_n| + |x_n - y| < 2\epsilon$, a contradiction. (b) With $\epsilon = 1$, $|x_n| \le |x| + 1$ for $n \ge N$, and finitely many earlier terms are bounded. ■

Theorem 1.7 (Limit laws)

If $x_n \to x$ and $y_n \to y$, then $x_n + y_n \to x + y$, $ax_n \to ax$, $x_n y_n \to xy$, and $x_n/y_n \to x/y$ when $y \neq 0$ (and $y_n \neq 0$). If $x_n \leq y_n$ for all large n, then $x \leq y$. If $x_n \leq z_n \leq y_n$ and $x = y$, then $z_n \to x$ (squeeze).

Proof. Products: $|x_n y_n - xy| \leq |x_n||y_n - y| + |y||x_n - x| \leq M|y_n - y| + |y||x_n - x|$, with M a bound for $|x_n|$. Quotients: it suffices that $1/y_n \to 1/y$; for large n, $|y_n| > |y|/2$, so $|1/y_n - 1/y| \leq 2|y_n - y|/|y|^2$. Order: if $x > y$, then $\epsilon = (x-y)/2$ gives $x_n > y_n$ for large n. Squeeze: $x - \epsilon < x_n \leq z_n \leq y_n < x + \epsilon$ for large n. ■

Example 1.1

(a) For $|a| < 1$, $a^n \to 0$: write $1/|a| = 1 + h$ with $h > 0$; by Bernoulli's inequality $(1+h)^n \geq 1 + nh$, so $|a|^n \leq 1/(1+nh) \to 0$.
(b) $n^{1/n} \to 1$: with $n^{1/n} = 1 + h_n$, the binomial theorem gives $n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)} \to 0$. Similarly $a^{1/n} \to 1$ for $a > 0$.
(c) If $x_n \to x$ then the averages $(x_1 + \cdots + x_n)/n \to x$ (Cesàro): split the sum at a large index N.

1.4 Monotone sequences, limsup and liminf

Theorem 1.8 (Monotone convergence)

A bounded monotone sequence of reals converges; an increasing one converges to $\sup_n x_n$.

Proof. Let (x_n) increase and $s = \sup\{x_n\}$ (completeness). Given ϵ, some $x_N > s - \epsilon$, and then $s - \epsilon < x_N \leq x_n \leq s$ for $n \geq N$. ■

In particular a series of nonnegative terms converges iff its partial sums are bounded.

Example 1.2 (The number e)

$a_n = (1 + 1/n)^n$ is increasing and bounded. By the binomial theorem $a_n = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j<k}(1 - \frac{j}{n})$; each term increases with n and there are more terms, so $a_n \uparrow$; and $a_n \leq \sum_{k=0}^{n} \frac{1}{k!} \leq 1 + \sum_{k \geq 1} 2^{1-k} = 3$. Its limit is e; comparing with the partial sums one finds $e = \sum_{k \geq 0} 1/k!$.

For a bounded sequence put $\overline{x}_n = \sup_{k \geq n} x_k$ (decreasing) and $\underline{x}_n = \inf_{k \geq n} x_k$ (increasing), and define

$$\limsup_{n \to \infty} x_n = \lim_{n} \sup_{k \geq n} x_k = \inf_{n} \sup_{k \geq n} x_k, \qquad \liminf_{n \to \infty} x_n = \sup_{n} \inf_{k \geq n} x_k.$$

These exist by Theorem 1.8, and $\liminf x_n \leq \limsup x_n$.

Proposition 1.9 (Characterization of limsup)

For a bounded sequence, $a = \limsup x_n$ iff for every $\epsilon > 0$: (i) $x_n < a + \epsilon$ for all sufficiently large n, and (ii) $x_n > a - \epsilon$ for infinitely many n. Dually for $\liminf$.

Proof. If $a = \inf_n \overline{x}_n$, some $\overline{x}_N < a + \epsilon$, giving (i); and every $\overline{x}_n \geq a$, so for each n some $k \geq n$ has $x_k > a - \epsilon$, giving (ii). Conversely (i) gives $\overline{x}_n \leq a + \epsilon$ for large n and (ii) gives $\overline{x}_n \geq a - \epsilon$ for all n, so $\lim \overline{x}_n = a$. ■

Theorem 1.10

A bounded sequence converges to a iff $\limsup x_n = \liminf x_n = a$.

Proof. If $x_n \to a$, then $a - \epsilon < x_k < a + \epsilon$ for $k \geq N$, so $a - \epsilon \leq \underline{x}_n \leq \overline{x}_n \leq a + \epsilon$ for $n \geq N$. Conversely $\underline{x}_n \leq x_n \leq \overline{x}_n$ and squeeze. ■

Thus "(i) for all large n" versus "(ii) for infinitely many n" is the essential difference between the two halves of the characterization. For unbounded sequences one allows the values $\pm\infty$.

Theorem 1.11 (Nested intervals)

If $I_n = [a_n, b_n]$ are bounded closed intervals with $I_1 \supset I_2 \supset \cdots$, then $\bigcap_n I_n \neq \emptyset$. If moreover $b_n - a_n \to 0$, the intersection is a single point.

Proof. (a_n) increases and is bounded by b_1; let $a = \lim a_n = \sup a_n$. For fixed m and $n \geq m$, $a_n \leq b_n \leq b_m$, so $a \leq b_m$; also $a \geq a_m$. Hence $a \in \bigcap I_m$. Two points in all I_n differ by at most $b_n - a_n \to 0$. ■

Theorem 1.8 is in fact equivalent to the completeness axiom (given the other axioms), and so is the nested interval theorem together with the Archimedean property.

1.5 Countable and uncountable sets

Sets A, B have the **same cardinality** if there is a bijection $A \to B$. A set is **countable** if it is finite or in bijection with $\mathbb{N}$.

Proposition 1.12

(a) A subset of a countable set is countable. (b) A countable union of countable sets is countable. (c) $\mathbb{N} \times \mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof. (a) A subset of $\mathbb{N}$ is finite or enumerated increasingly. (c) $(m, n) \mapsto 2^m 3^n$ injects $\mathbb{N} \times \mathbb{N}$ into $\mathbb{N}$; $\mathbb{Z}$ is enumerated as $0, 1, -1, 2, -2, \ldots$; $\mathbb{Q}$ is the image of $\mathbb{Z} \times \mathbb{N}$ under $(p, q) \mapsto p/q$. (b) If $A_m = \{a_{m1}, a_{m2}, \ldots\}$, then $(m, n) \mapsto a_{mn}$ maps (a subset of) $\mathbb{N} \times \mathbb{N}$ onto $\bigcup A_m$; choose for each element the least preimage. ■

Theorem 1.13 (Cantor)

$\mathbb{R}$ is uncountable. In fact no interval $[a, b]$ with $a < b$ is countable.

Proof. Let $x_1, x_2, \ldots$ be any sequence in $[a, b]$. Divide $[a, b]$ into three closed thirds and choose one, I_1, not containing x_1; divide I_1 into thirds and choose $I_2 \subset I_1$ not containing x_2, and so on. By Theorem 1.11 some $x \in \bigcap I_n$, and $x \neq x_n$ for every n. So no sequence exhausts $[a, b]$. ■

Example 1.3

(a) The irrationals are uncountable (otherwise $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be countable), although both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense.
(b) The set of all subsets of $\mathbb{N}$ is uncountable (Cantor's diagonal argument: no $f : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ is onto, since $\{n : n \notin f(n)\}$ is missed). Writing reals in $[0, 1)$ in binary shows that $[0, 1)$ has the cardinality of $\mathcal{P}(\mathbb{N})$.
(c) The **Cantor set** $K = \bigcap K_n$, where $K_0 = [0, 1]$ and K_{n+1} removes the open middle third of every interval of K_n, has total removed length $\sum 2^n/3^{n+1} = 1$ but is uncountable: it consists

of the numbers whose ternary expansions use only the digits 0 and 2.